

Preparing for the AWS Cloud Practitioner Exam: An Alphabetical Guide to AWS Services and Key Features

To effectively prepare for the AWS Cloud Practitioner exam, it is crucial to understand the domains covered. The examination is structured around four key domains: Cloud Concepts, Security & Compliance, Cloud Technology and Services, and Billing, Pricing and Support.³ While this report primarily focuses on services relevant to the "**Cloud Technology and Services**" domain, the descriptions will also touch upon aspects related to "**Billing and Pricing**" and "**Security & Compliance**" where pertinent. The official AWS Certified Cloud Practitioner Exam Guide stands as the definitive resource for comprehending the complete scope of the exam and identifying the specific services that are considered in scope.²

The AWS Cloud encompasses a vast ecosystem of services, which can be broadly categorized to facilitate understanding. Some of the major categories that are particularly relevant for the Cloud Practitioner exam include **Compute, Storage, Database, Networking & Content Delivery, Security, Identity, & Compliance, Management & Governance, Analytics, and Serverless**.⁵

Core AWS Service Categories and Examples

Category	Example Services
Compute	EC2, Lambda, Elastic Beanstalk
Storage	S3, EBS, EFS, Glacier
Database	RDS, DynamoDB, Aurora
Networking & Content Delivery	VPC, CloudFront, Route 53
Security, Identity, & Compliance	IAM, KMS, Shield
Management & Governance	CloudWatch, CloudTrail, Config, Trusted Advisor
Analytics	Athena, EMR, QuickSight
Serverless	Lambda, API Gateway, S3

Alphabetical List of AWS Services and Key Features

- **AWS Amplify**¹
 - A set of **tools and services** designed to help **frontend web and mobile developers** build and deploy **scalable full-stack applications** powered by AWS.
 - Enables developers to easily **configure backends** and **connect their applications to AWS services**.
- **Amazon API Gateway**¹
 - A **fully managed service** that simplifies the **creation, publication, maintenance, monitoring, and security of APIs** at any scale.
 - Supports the creation of **RESTful APIs** and **WebSocket APIs**.
 - Handles essential API management tasks such as **traffic management, authorization, access control, and version control**.
 - Operates on a **pay-as-you-go pricing model**. This service is fundamental for modern application development, allowing for the creation of scalable and secure interfaces. Its frequent appearance in Cloud Practitioner resources underscores its importance.
- **Amazon AppStream 2.0**¹
 - A **fully managed application streaming service** that allows users to **securely access desktop applications** from any location.
 - Provides a **centralized platform for managing desktop applications**, which are then **streamed to users' devices**.
- **Amazon Athena**¹
 - An **interactive query service** that is **serverless**, making it easy to **analyze data directly in Amazon S3** using standard SQL.
 - Eliminates the need to manage infrastructure or data warehouses for querying data in S3.
- **Amazon Aurora**¹
 - A **fully managed relational database service** that is **compatible with MySQL and PostgreSQL**.
 - Designed for the cloud, offering **enhanced performance and availability** compared to traditional databases, along with the **simplicity and cost-effectiveness of open-source databases**.
- **AWS Auto Scaling**¹
 - A service that **monitors applications** and **automatically adjusts the compute capacity** to maintain predictable performance at the lowest possible cost.

- Enables **dynamic scaling of EC2 instances** and other resources based on defined metrics and policies.
- **AWS Backup**¹
 - A **fully managed service** that **centralizes and automates the backup of data** across various AWS services in the cloud and on-premises.
 - Provides a **consistent way to manage backup policies** and **monitor backup activity**.
- **AWS Batch**¹
 - A **fully managed batch processing service** that allows developers, scientists, and engineers to easily run **hundreds of thousands of batch computing jobs** on AWS.
 - Efficiently manages compute resources based on the volume and specific resource requirements of batch jobs.
- **AWS Budgets**¹
 - A service that enables users to **set custom budgets to track their AWS costs and usage**, providing alerts when costs or usage exceed defined thresholds.
 - Helps in **managing and controlling AWS spending**.
- **Amazon CloudFront**¹
 - A **fast content delivery network (CDN) service** that **securely delivers data, videos, applications, and APIs** to customers globally with **low latency** and **high transfer speeds**.
 - **Caches content at edge locations** around the world, reducing the load on origin servers and improving user experience.
- **AWS CloudFormation**¹
 - An **infrastructure-as-code service** that allows users to **define and provision AWS infrastructure resources** in a **declarative way using templates**.
 - Enables the **automation and consistent deployment of entire AWS environments**.
- **Amazon CloudWatch**¹
 - A **monitoring and observability service** that provides **data and actionable insights** to **monitor applications, respond to performance changes, optimize resource utilization, and gain a unified view of operational health**.
 - Offers features such as **metrics, logs, and alarms** for monitoring AWS resources and applications.

- **AWS CodeBuild** ¹
 - A **fully managed continuous integration service** that **compiles source code, runs tests, and produces software packages** that are ready to be deployed.
- **AWS CodeCommit** ¹
 - A **fully managed source control service** that hosts **secure Git-based repositories**.
- **AWS CodeDeploy** ¹
 - A **fully managed deployment service** that **automates code deployments** to various compute services, including **EC2 instances** and **on-premises servers**.
- **Amazon CodeGuru** ¹⁴
 - A **machine learning service** that helps developers **improve code quality and identify expensive lines of code**.¹⁶
 - **CodeGuru Reviewer** automates code reviews to detect issues like **security vulnerabilities, resource leaks, and deviations from best practices**.¹⁸ It integrates with code repositories like **GitHub** and **AWS CodeCommit**.¹⁸
 - **CodeGuru Profiler** helps identify an application's **most expensive lines of code**, understand **runtime behavior**, and get recommendations to **improve performance and reduce costs**.¹⁴ It **continuously profiles applications in production** with minimal overhead.¹⁵
- **Amazon Cognito** ¹
 - A service that allows developers to easily add **user sign-up, sign-in, and access control** to their web and mobile applications.
- **Amazon Comprehend** ¹
 - A **natural language processing (NLP) service** that uses **machine learning** to extract insights and relationships from text.
- **Amazon Connect** ¹
 - An **easy-to-use, omnichannel cloud contact center service** that helps organizations provide **superior customer service at a lower cost**.
- **AWS Cost Explorer** ¹
 - An **intuitive interface** that allows users to **visualize, understand, and manage their AWS costs and usage over time**.
- **AWS Data Exchange** ¹
 - A service that simplifies the process of **finding, subscribing to, and using third-party data in the cloud**.
- **AWS Direct Connect** ¹
 - A **cloud service solution** that facilitates the establishment of a **dedicated network connection from on-premises environments to AWS**.

- **Amazon DynamoDB**¹
 - A **fully managed NoSQL key-value and document database** that delivers **single-digit millisecond performance at any scale**.
- **Amazon EC2 (Elastic Compute Cloud)**¹
 - A **web service** that provides **resizable compute capacity in the cloud** in the form of **virtual servers known as instances**.
 - Offers a **wide range of instance types** optimized for different workloads.
- **Amazon ECR (Elastic Container Registry)**¹
 - A **fully managed Docker container registry** that simplifies the **storage, management, and deployment of Docker container images**.
- **Amazon ECS (Elastic Container Service)**¹
 - A **fully managed container orchestration service** that makes it easy to **deploy, manage, and scale Docker containers**.
- **Amazon EFS (Elastic File System)**¹
 - A **fully managed, scalable, elastic, and serverless Network File System (NFS) file system** for use with AWS Cloud services and on-premises resources.
 - Allows **multiple EC2 instances to access shared data concurrently**.
- **AWS Elastic Beanstalk**¹
 - An **easy-to-use service** for **deploying and scaling web applications and services** developed with various programming languages and Docker on familiar server environments.
 - **Automates the deployment and management of the underlying infrastructure**.
- **Elastic Load Balancing (ELB)**¹
 - A service that **automatically distributes incoming application traffic** across multiple targets, such as **EC2 instances, containers, IP addresses, and Lambda functions**.
 - Offers different types of load balancers (**Application Load Balancer, Network Load Balancer, and Classic Load Balancer**) to suit various application needs.
- **Amazon EMR (Elastic MapReduce)**¹
 - A **managed big data service** that provides a **cost-effective way to process vast amounts of data** using open-source frameworks such as **Apache Hadoop and Apache Spark**.
- **Amazon EventBridge**¹
 - A **serverless event bus** that enables the **easy connection of applications** using data from various sources, including custom applications, SaaS applications, and AWS services.

- **AWS Fargate**¹
 - A **serverless, pay-as-you-go compute engine** that allows users to **run containers without the need to manage underlying servers or clusters**.
 - Works with both **Amazon ECS** and **Amazon EKS**.
- **Amazon FSx**¹
 - Provides **fully managed third-party file systems**, including **Amazon FSx for Windows File Server**, **Amazon FSx for NetApp ONTAP**, and **Amazon FSx for Lustre**.
- **AWS Glue**¹
 - A **fully managed extract, transform, and load (ETL) service** that simplifies the **preparation and loading of data for analytics**.
- **AWS Global Accelerator**¹
 - A **networking service** that enhances the **availability and performance of user traffic** by routing it through the AWS global network infrastructure.
- **AWS IAM (Identity and Access Management)**¹
 - A **web service** that enables **secure control of access to AWS services and resources**.
 - Allows the **creation and management of AWS users and groups**, and the use of **permissions to grant or deny access to resources**.
- **AWS IoT Core**¹
 - A **managed cloud service** that enables **connected devices to easily and securely interact with cloud applications and other devices**.
- **Amazon Kinesis**¹
 - A **platform for streaming data on AWS**, offering services for **real-time data ingestion, processing, and delivery**, including **Kinesis Data Streams**, **Kinesis Data Firehose**, **Kinesis Data Analytics**, and **Kinesis Video Streams**.
- **AWS KMS (Key Management Service)**¹
 - A **managed service** that simplifies the **creation and management of encryption keys**, controlling their use across a wide range of AWS services and in applications.
- **AWS Lambda**¹
 - A **serverless, event-driven compute service** that allows users to **run code without provisioning or managing servers**.
 - Code is executed only when triggered by specific events.
- **Amazon Lex**¹
 - A service for **building conversational interfaces** into applications using voice and text.

- **Amazon Lightsail**¹
 - An **easy-to-use cloud platform** that offers **virtual private servers (VPS), databases, containers, and storage** at a predictable monthly price.
- **Amazon Macie**¹
 - A **fully managed data security and data privacy service** that uses **machine learning and pattern matching** to discover and protect sensitive data in AWS.
- **AWS Management Console**¹
 - A **web-based interface** for **accessing and managing AWS services**.
- **AWS Marketplace**¹
 - A **digital catalog** featuring thousands of software listings from independent software vendors that can be easily found, tested, bought, and deployed on AWS.
- **Amazon Neptune**¹
 - A **fully managed graph database service**.
- **Amazon Polly**¹
 - A service that **converts text into lifelike speech**, enabling the creation of speech-enabled applications.
- **Amazon QuickSight**¹
 - A **fast, cloud-powered business intelligence service** that makes it easy to **deliver insights to everyone within an organization** through interactive dashboards.
- **Amazon RDS (Relational Database Service)**¹
 - A **managed service** that simplifies the **setup, operation, and scaling of relational databases in the cloud**.
 - Supports various database engines, including **Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and SQL Server**.
- **Amazon Redshift**¹
 - A **fast, fully managed, petabyte-scale data warehouse service**.
- **Amazon Rekognition**¹
 - A service that makes it easy to add **image and video analysis capabilities** to applications using proven, highly scalable **deep learning technology**.
- **Amazon Route 53**¹
 - A **highly available and scalable cloud Domain Name System (DNS) web service**.

- **Amazon S3 (Simple Storage Service)** ¹
 - An **object storage service** offering industry-leading **scalability, data availability, security, and performance**.
 - Provides **various storage classes** optimized for different access patterns and cost requirements.
- **Amazon S3 Glacier** ¹
 - A **secure, durable, and extremely low-cost Amazon S3 cloud storage class** designed for **data archiving and long-term backup**.
- **Amazon SageMaker** ¹
 - A **fully managed machine learning service** that provides every developer and data scientist with the ability to **build, train, and deploy machine learning models quickly**.
- **AWS Secrets Manager** ¹
 - A service that helps **securely store and retrieve secrets** such as passwords, API keys, and other sensitive information.
- **Amazon SES (Simple Email Service)** ¹
 - A **cost-effective email service** built to help digital marketers and application developers send **marketing, notification, and transactional emails**.
- **AWS Shield** ¹
 - A **managed Distributed Denial of Service (DDoS) protection service** that safeguards applications running on AWS.
- **Amazon SNS (Simple Notification Service)** ¹
 - A **highly available, durable, secure, fully managed pub/sub messaging service** that enables the **decoupling of microservices, distributed systems, and serverless applications**.
- **AWS Snowball** ²³
 - A service that provides **secure, rugged devices to transfer large amounts of data** into and out of AWS.²³ It's particularly useful when **network connectivity is limited or cost-prohibitive**.²⁵
 - Offers different device types like **Snowball Edge Storage Optimized** for large-scale data migration and **Snowball Edge Compute Optimized** with computing capabilities for edge processing.²⁶
 - Features include **high-speed data transfer, 256-bit encryption, tamper-resistant design, and integration with Amazon S3**.²⁷ Data on the device is **securely erased** after the transfer is complete.²⁷

- **Amazon SQS (Simple Queue Service)** ¹
 - A **fully managed message queuing service** that enables the **decoupling and scaling of microservices, distributed systems, and serverless applications**.
- **AWS Step Functions** ¹
 - A **serverless orchestration service** that allows users to **coordinate multiple AWS services into serverless workflows** using visual workflows.
- **AWS Storage Gateway** ¹
 - A **hybrid cloud storage service** that provides **on-premises access to virtually unlimited cloud storage**, simplifying storage management and reducing costs for key hybrid cloud use cases.
- **AWS Systems Manager** ¹
 - A **secure and scalable management service** that provides a **unified interface for automating operational tasks** across AWS resources.
- **Amazon Textract** ¹
 - A **fully managed machine learning service** that **automatically extracts text and data from scanned documents**.
- **Amazon Transcribe** ¹
 - An **automatic speech recognition (ASR) service** that makes it easy for developers to add **speech-to-text capabilities** to their applications.
- **Amazon Translate** ¹
 - A **neural machine translation service** that delivers **fast, high-quality, and affordable language translation**.
- **AWS Trusted Advisor** ¹
 - An **online tool** that provides **real-time guidance** to help provision resources following AWS best practices in the areas of **cost optimization, security, fault tolerance, service limits, and performance**.
- **Amazon VPC (Virtual Private Cloud)** ¹
 - A service that enables the **provisioning of a logically isolated section of the AWS Cloud** where AWS resources can be launched in a **virtual network defined by the user**.
- **AWS WAF (Web Application Firewall)** ¹
 - A **web application firewall** that helps **protect web applications from common web exploits and bots** that could affect application availability, compromise security, or consume excessive resources.

- **Amazon WorkSpaces** ¹
 - A **fully managed, secure Desktop-as-a-Service (DaaS) solution** that allows the provisioning of **Windows or Linux desktops in the cloud**.
- **AWS X-Ray** ¹
 - A service that helps developers **analyze and debug distributed applications** in production or under development, such as those built using a microservices architecture.